

Nonparametric Statistics

Chapter 2: Density Estimation

Hoang Van Ha

hvha@hcmus.edu.vn

Faculty of Mathematics and Computer Science
University of Science, VNU-HCM

Outline

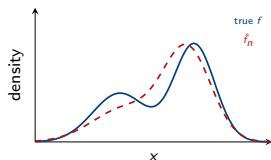
- 1 Introduction to Density Estimation
- 2 Risk, Bias–Variance Tradeoff, and Smoothing Parameter
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Motivation: What is Density Estimation?

- **Data:** $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} F$, where F has *unknown* density $f = F'$.
- **Goal:** Estimate f from observations with *as few assumptions as possible*.
- We typically assume $f \in \mathcal{F}$, a *function space* (Hölder, Sobolev, ...).
- **Why?** Knowledge of f reveals:
 - ▶ Number and location of *modes*
 - ▶ Regions of high/low density
 - ▶ Skewness, tail behavior, ...

Key Question

Given $X_1, \dots, X_n$, how do we best *reconstruct* the shape of the underlying distribution?



Parametric vs. Nonparametric Estimation

Parametric Approach

Assume $f(\cdot; \theta)$ is known up to $\theta \in \Theta \subset \mathbb{R}^k$.

Pros: Efficient if model is correct.

Cons: Model misspecification leads to **biased** inference.

Nonparametric Approach

Let $f \in \mathcal{F}$, an infinite-dimensional function space.

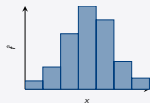
Pros: **Flexible**, no strong model assumptions.

Cons: Slower convergence rates.

- **Notation:** $\mathbb{E}_f[\cdot]$ denotes expectation when the data density is f .
- The estimator $\hat{f}_n$ will depend on a **smoothing parameter** (bandwidth h , number of bins m , projection dimension D , ...).
- Choosing this parameter is the central challenge in density estimation.

Three Main Estimators in This Chapter

1. Histogram



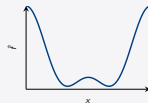
Piecewise constant; bins of width h

2. Kernel (KDE)



Smooth; bandwidth h

3. Projection



Series expansion; dimension D

All three estimators depend on a **smoothing parameter** λ ($= h$ or D). The optimal choice involves a *bias-variance tradeoff*.

Loss Functions and Risk

Definition 1 (Risk Function)

For an estimator $\hat{f}_n$, the **risk** is $R(\hat{f}_n, f) = \mathbb{E}_f[d(\hat{f}_n, f)]$ where d is a loss function.

Common loss functions:

MISE (Mean Integrated Squared Error):

$$R(\hat{f}_n, f) = \mathbb{E} \int (\hat{f}_n(x) - f(x))^2 dx$$

Global quality measure.

Pointwise MSE at x_0 :

$$R_{x_0}(\hat{f}_n, f) = \mathbb{E} \left[(\hat{f}_n(x_0) - f(x_0))^2 \right]$$

Local quality measure.

- Both risks depend on the (unknown) true density f .
- We seek estimators that *minimize* risk over a function class $\mathcal{F}$.

Bias–Variance Decomposition

For any estimator $\hat{f}_n(x)$ at a fixed point x :

$$\text{MSE}(\hat{f}_n(x)) = \underbrace{\left(\mathbb{E}[\hat{f}_n(x)] - f(x)\right)^2}_{\text{Bias}^2} + \underbrace{\text{Var}(\hat{f}_n(x))}_{\text{Variance}}$$

Bias

Systematic error from using a simplified/smoothed model.

- Decreases as smoothing $\downarrow$ (more complex model)
- Independent of n
- $\propto h^2$ (or $D^{-2\beta}$ for projection)

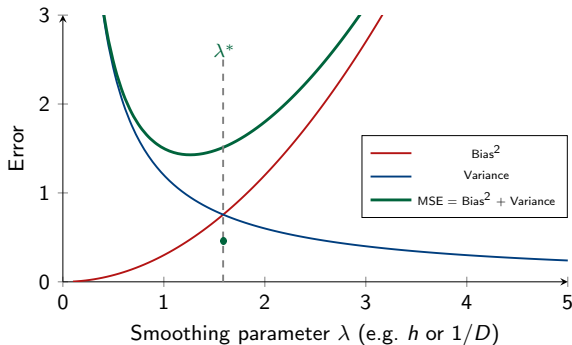
Variance

Statistical fluctuation due to finite sample.

- Increases as smoothing $\downarrow$
- Decreases as $n \uparrow$
- $\propto 1/(nh)$ (or D/n)

Key tradeoff: Less smoothing $\Rightarrow$ bias $\downarrow$ but variance $\uparrow$. More smoothing $\Rightarrow$ bias $\uparrow$ but variance $\downarrow$.

The Bias–Variance Tradeoff: Visualization



The **optimal** λ^* minimizes MSE, balancing bias and variance. It depends on the **unknown** smoothness of f .

Smoothness Classes for Density Functions

To quantify bias, we assume f belongs to a **function space** parameterized by *smoothness* $\beta > 0$ and *radius* $L > 0$.

Hölder Class $\mathcal{H}(\beta, L)$

Let $r = \lfloor \beta \rfloor$. Then

$$\mathcal{H}(\beta, L) = \left\{ f \in C^r : |f^{(r)}(x) - f^{(r)}(y)| \leq L|x - y|^{\beta-r}, \forall x, y \right\}.$$

Sobolev Class $\mathcal{S}(\beta, L)$

$$\mathcal{S}(\beta, L) = \left\{ f \in L^2 : \int |f^*(u)|^2 (1 + |u|^2)^\beta du \leq L^2 \right\},$$

where f^* is the Fourier transform of f .

- **Larger** $\beta \Rightarrow$ smoother $f \Rightarrow$ *faster* convergence rates.
- The parameter β is typically **unknown** in practice.

Optimal Convergence Rates

Proposition 1 (Minimax lower bound (Tsybakov 2009))

For any estimator $\hat{f}_n$ and $f \in \mathcal{H}(\beta, L)$ with β, L known:

$$\sup_{f \in \mathcal{H}(\beta, L)} \mathbb{E} \int (\hat{f}_n(x) - f(x))^2 dx \geq C_{\beta, L} n^{-2\beta/(2\beta+1)}.$$

Optimal rate of convergence: $n^{-2\beta/(2\beta+1)}$

- $\beta = 1$ (Lipschitz): rate $n^{-2/3}$ (histogram achieves this)
- $\beta = 2$ (twice diff.): rate $n^{-4/5}$ (kernel / projection achieves this)
- $\beta \rightarrow \infty$ (analytic): rate $\rightarrow n^{-1}$ (parametric rate)

Remark 1

The optimal smoothing parameter satisfies $\lambda^* \propto n^{-1/(2\beta+1)}$, which *depends on the unknown β* . This motivates data-driven selection.

Bandwidth/Parameter Selection: Cross-Validation

Cross-validation (CV) is the standard data-driven method for selecting λ .

Key idea: The loss $L(h) = \int (\hat{f}_n(x) - f(x))^2 dx$ can be written as

$$L(h) = \int \hat{f}_n^2(x) dx - 2 \int \hat{f}_n(x)f(x) dx + \underbrace{\int f^2(x) dx}_{\text{const.}}$$

Define the **risk proxy**:

$$J(h) = \int \hat{f}_n^2(x) dx - 2 \int \hat{f}_n(x)f(x) dx$$

Leave-one-out CV score:

$$\hat{J}(h) = \int \hat{f}_n^2(x) dx - \frac{2}{n} \sum_{i=1}^n \hat{f}_n^{(-i)}(X_i),$$

where $\hat{f}_n^{(-i)}$ is the estimator computed without observation X_i .

Choose: $\hat{h} = \operatorname{argmin}_h \hat{J}(h)$

- CV is *asymptotically optimal*: $I_{CV}/I^* \rightarrow 1$ a.s. (Stone 1984).

The Histogram Estimator: Definition

Setup: Suppose f has support $[0, 1]$. Let m be a positive integer.

Bins: Partition $[0, 1]$ into m equal bins of width $h = 1/m$:

$$B_j = \left[\frac{j-1}{m}, \frac{j}{m} \right), \quad j = 1, \dots, m.$$

Counts: Let $Y_j = \#\{i : X_i \in B_j\}$ and $\hat{p}_j = Y_j/n$.

Histogram estimator:

$$\hat{f}_n(x) = \sum_{j=1}^m \frac{\hat{p}_j}{h} \mathbf{1}(x \in B_j) = \frac{1}{nh} \sum_{j=1}^m Y_j \mathbf{1}(x \in B_j)$$

- Interpretation: *fraction of observations in bin B_j , divided by bin width.*
- $\hat{f}_n \geq 0$ and $\int_0^1 \hat{f}_n(x) dx = 1$: it is a **valid density**.
- Smoothing parameter: binwidth $h = 1/m$ (or equivalently, m bins).

Histogram: Bias Analysis

For $x \in B_j$ and the true probability $p_j = \int_{B_j} f(u) du$:

$$\mathbb{E}[\hat{f}_n(x)] = \frac{p_j}{h} = \frac{1}{h} \int_{B_j} f(u) du.$$

Proposition 2 (Histogram bias)

For $x \in B_j$ and $f \in \mathcal{C}^1$:

$$\mathbb{E}[\hat{f}_n(x)] - f(x) = f'(x)(h(j - \frac{1}{2}) - x) + O(h^2),$$

and the **integrated squared bias** satisfies:

$$\int_0^1 b^2(x) dx = \frac{h^2}{12} \int_0^1 (f'(x))^2 dx + o(h^2).$$

The bias reflects how much f varies *within* each bin.
 $h \rightarrow 0$ (more bins) $\Rightarrow$ smaller bias; bins better approximate f .

Histogram: Variance Analysis

$(Y_1, \dots, Y_m)$ follows a *Multinomial* $(n, p_1, \dots, p_m)$ distribution.

So $Y_j \sim \text{Bin}(n, p_j)$, giving:

$$\mathbb{V}\text{ar}(\hat{f}_n(x)) = \frac{p_j(1-p_j)}{nh^2} \approx \frac{f(x)}{nh}, \quad x \in B_j.$$

Integrating:

Integrated variance:

$$\int_0^1 \mathbb{V}\text{ar}(\hat{f}_n(x)) dx \approx \frac{1}{nh} \int_0^1 f(x) dx = \frac{1}{nh}.$$

- Variance $\uparrow$ as $h \downarrow$ (more bins = noisier estimate).
- Variance $\downarrow$ as $n \uparrow$.

Histogram: MISE and Optimal Binwidth

Theorem 1 (MISE of Histogram (Wasserman 2006, Ch. 6))

Under mild regularity conditions:

$$\text{MISE}(\hat{f}_n, f) = \frac{h^2}{12} \int_0^1 (f'(x))^2 dx + \frac{1}{nh} + O\left(h^3 + \frac{1}{n}\right).$$

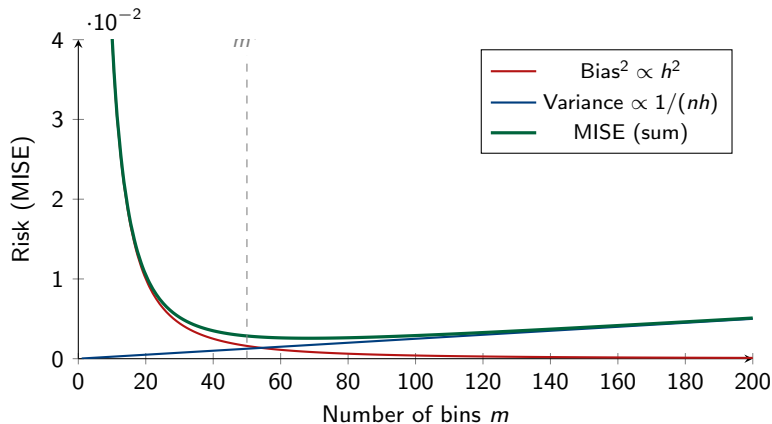
Optimal binwidth: Minimize over h :

$$h^* = \operatorname{argmin}_h \left\{ \frac{h^2}{12} \int_0^1 (f')^2 + \frac{1}{nh} \right\} \Rightarrow h^* = \left(\frac{6}{\int_0^1 (f'(x))^2 dx \cdot n} \right)^{1/3}$$

Consequence: $h^* \propto n^{-1/3}$ and $\text{MISE}(\hat{f}_n, f)|_{h=h^*} \propto n^{-2/3}$.

This is the **minimax optimal rate** for $f \in \mathcal{H}(1, L)$.

Histogram: Effect of Binwidth



Too few bins (m small, h large):
 $\Rightarrow$ oversmoothing, high bias.

Too many bins (m large, h small):
 $\Rightarrow$ undersmoothing, high variance.

Histogram: Cross-Validation for Binwidth Selection

The optimal h^* depends on $\int (f')^2$, which is **unknown**.

Leave-one-out CV for histograms:

$$\hat{J}(h) = \frac{2}{(n-1)h} - \frac{n+1}{(n-1)h} \sum_{j=1}^m \hat{p}_j^2,$$

and we choose $\hat{m} = \operatorname{argmin}_m \hat{J}(1/m)$.

Simplified form:

$$\hat{J}(h) = \frac{2}{(n-1)h} - \frac{n+1}{(n-1)h} \sum_{j=1}^m \left(\frac{Y_j}{n} \right)^2$$

Choose $\hat{m}$ that minimizes this score over a grid of m values.

- Result: $\hat{J}$ is an *unbiased* estimator of $J(h)$ up to a constant.
- CV-selected $\hat{m}$ achieves the **same convergence rate** as oracle m^* .

Histogram: Limitations

Drawbacks of Histograms

- ❶ **Discontinuous:** $\hat{f}_n$ is a step function—cannot model smooth densities.
- ❷ **Sensitive to bin origin:** shifting the grid changes the shape significantly.
- ❸ **Suboptimal rate:** $\text{MISE} \propto n^{-2/3}$ vs. $n^{-4/5}$ for kernel estimators.
- ❹ **Poor in high dimensions:** The curse of dimensionality is severe.

→ These limitations motivate **kernel density estimation (KDE)**, which produces *smooth* estimates and achieves the optimal $n^{-4/5}$ rate.

Kernel Density Estimator: Definition

Motivation: Replace the hard bin indicator by a smooth weight function.

Definition 2 (Kernel)

A function $K : \mathbb{R} \rightarrow \mathbb{R}$ is a **kernel** if:

$$\int K(u) du = 1, \quad \int u K(u) du = 0, \quad \sigma_K^2 = \int u^2 K(u) du < \infty.$$

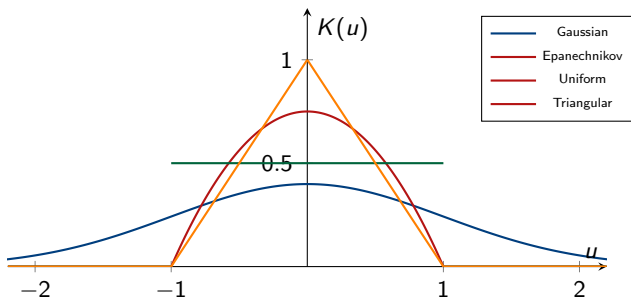
Kernel Density Estimator (KDE):

$$\hat{f}_n(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - X_i}{h}\right) = \frac{1}{n} \sum_{i=1}^n K_h(x - X_i)$$

where $K_h(u) = h^{-1}K(u/h)$ and $h > 0$ is the **bandwidth**.

- Interpretation: *place a scaled kernel K_h at each data point X_i , then average.*
- $\hat{f}_n \geq 0$ and $\int \hat{f}_n = 1$ whenever K is a density.

Common Kernel Functions



Epanechnikov kernel: $K(u) = \frac{3}{4}(1 - u^2)\mathbf{1}(|u| \leq 1)$ is **MSE-optimal** among all kernels with compact support (minimizes $\int K^2 \cdot \sigma_K^4$). The choice of K has *much less impact* than the choice of h .

KDE: Bias and Variance

Theorem 2 (Pointwise MSE of KDE (Wasserman 2006, Thm. 6.28))

Assume f is absolutely continuous, $f'' \in L^2$, and K satisfies the kernel conditions. Then:

$$\mathbb{E}[\hat{f}_n(x)] - f(x) = \frac{1}{2} \sigma_K^2 h^2 f''(x) + O(h^4),$$

$$\mathbb{V}\text{ar}(\hat{f}_n(x)) = \frac{f(x) \int K^2(u) du}{nh} + O\left(\frac{1}{n}\right).$$

Pointwise MSE (leading terms):

$$\text{MSE}(\hat{f}_n(x)) \approx \underbrace{\frac{\sigma_K^4 h^4 (f''(x))^2}{4}}_{\text{Bias}^2 \propto h^4} + \underbrace{\frac{f(x) \int K^2 du}{nh}}_{\text{Variance} \propto (nh)^{-1}}$$

Hölder Class and Kernel of Order ℓ

Definition 3 (Hölder Class $\Sigma(\beta, L)$)
(Tsybakov, Def. 1.2))

Let $\ell = \lfloor \beta \rfloor$. The **Hölder class** $\Sigma(\beta, L)$ on $\mathbb{R}$ is the set of ℓ -times differentiable functions f such that:

$$|f^{(\ell)}(x) - f^{(\ell)}(x')| \leq L|x - x'|^{\beta - \ell}, \quad \forall x, x' \in \mathbb{R}.$$

Definition 5 (Kernel of order ℓ)
(Tsybakov, Def. 1.3))

$K : \mathbb{R} \rightarrow \mathbb{R}$ is a **kernel of order ℓ** if the functions $u \mapsto u^j K(u)$ are integrable and:

$$\int K(u) du = 1, \quad \int u^j K(u) du = 0, \quad j = 1, \dots, \ell$$

Definition 4 (Density class $P(\beta, L)$)

$$P(\beta, L) = \{p \geq 0 : \int p = 1, p \in \Sigma(\beta, L)\}.$$

Interpretation:

A kernel of order ℓ cancels all polynomial bias terms up to degree ℓ .

Order $\ell = 1$: standard kernels (Gaussian, Epanechnikov, ...)

Order $\ell \geq 2$: *higher-order kernels* (may take negative values).

Variance Bound

Proposition 3 (Variance bound Tsybakov (2009), Prop. 1.1)

Suppose p satisfies $p(x) \leq p_{\max} < \infty$ for all x . Let $K : \mathbb{R} \rightarrow \mathbb{R}$ with $\int K^2(u) du < \infty$. Then for any $x_0 \in \mathbb{R}$, $h > 0$, and $n \geq 1$:

$$\sigma^2(x_0) := \text{Var}(\hat{p}_n(x_0)) \leq \frac{C_1}{nh}, \quad C_1 = p_{\max} \int K^2(u) du.$$

Proof sketch. Write $\eta_i(x_0) = K\left(\frac{X_i - x_0}{h}\right) - \mathbb{E}\left[K\left(\frac{X_i - x_0}{h}\right)\right]$. The η_i are i.i.d., zero-mean, and:

$$\mathbb{E}[\eta_i^2] \leq \mathbb{E}\left[K^2\left(\frac{X_i - x_0}{h}\right)\right] = \int K^2\left(\frac{z - x_0}{h}\right) p(z) dz \leq p_{\max} h \int K^2(u) du.$$

Since $\hat{p}_n(x_0) = \frac{1}{nh} \sum_i \left(K\left(\frac{X_i - x_0}{h}\right)\right)$, independence gives: $\sigma^2(x_0) = \frac{1}{n^2 h^2} \sum_i \mathbb{E}[\eta_i^2] \leq \frac{C_1}{nh}$.

Key consequence: $nh \rightarrow \infty \Rightarrow \sigma^2(x_0) \rightarrow 0$. Variance control requires **bandwidth** $h \gg 1/n$.

Bias Bound

Proposition 4 (Bias bound Tsybakov (2009), Prop. 1.2)

Let $p \in P(\beta, L)$ and K be a kernel of order $\ell = \lfloor \beta \rfloor$ with $\int |u|^\beta |K(u)| du < \infty$. Then for all $x_0 \in \mathbb{R}$, $h > 0$, $n \geq 1$:

$$|b(x_0)| := |\mathbb{E}[\hat{p}_n(x_0)] - p(x_0)| \leq C_2 h^\beta, \quad C_2 = \frac{L}{\ell!} \int |u|^\beta |K(u)| du.$$

Proof sketch. By the change of variable $u = (z - x_0)/h$:

$$b(x_0) = \int K(u) [p(x_0 + uh) - p(x_0)] du.$$

Taylor expansion to order ℓ with $p \in \Sigma(\beta, L)$:

$$p(x_0 + uh) = p(x_0) + p'(x_0)uh + \cdots + \frac{(uh)^\ell}{\ell!} p^{(\ell)}(x_0 + \tau uh).$$

Since K has order ℓ (i.e. $\int u^j K(u) du = 0$ for $j = 1, \dots, \ell$), the first ℓ terms vanish, leaving:

$$|b(x_0)| \leq \int |K(u)| \frac{|uh|^\ell}{\ell!} |p^{(\ell)}(x_0 + \tau uh) - p^{(\ell)}(x_0)| du \leq C_2 h^\beta.$$

Key consequence: $h \rightarrow 0 \Rightarrow |b(x_0)| \rightarrow 0$. Higher-order kernel ($\ell \nearrow$) with smoother p ($\beta \nearrow$) gives **faster bias decay** h^β .

Combining Prop. 1.1 & 1.2: MSE Bound and Optimal h

From Propositions 1.1 and 1.2, the MSE at x_0 satisfies:

$$\text{MSE}(\hat{p}_n(x_0)) = b^2(x_0) + \sigma^2(x_0) \leq \underbrace{C_2^2 h^{2\beta}}_{\text{Bias}^2 \nearrow h} + \underbrace{\frac{C_1}{nh}}_{\text{Variance} \searrow h}$$

Optimal bandwidth: Minimize the bound over $h > 0$:

$$\frac{d}{dh} \left[C_2^2 h^{2\beta} + \frac{C_1}{nh} \right] = 0 \Rightarrow h_n^* = \left(\frac{C_1}{2\beta C_2^2} \right)^{\frac{1}{2\beta+1}} n^{-\frac{1}{2\beta+1}}$$

Theorem 3 (Tsybakov (2009), Thm. 1.1)

Under the assumptions of Propositions 1.1 and 1.2, fix $\alpha > 0$ and set $h = \alpha n^{-1/(2\beta+1)}$. Then for all $n \geq 1$:

$$\sup_{x_0 \in \mathbb{R}} \sup_{p \in P(\beta, L)} \mathbb{E}_p \left[(\hat{p}_n(x_0) - p(x_0))^2 \right] \leq C n^{-\frac{2\beta}{2\beta+1}},$$

where $C > 0$ depends only on β , L , α , and K .

KDE: Optimal Bandwidth and MISE

Integrating the pointwise MSE:

$$\text{MISE}(\hat{f}_n, f) \approx \frac{\sigma_K^4 h^4}{4} \int (f''(x))^2 dx + \frac{\int K^2(u) du}{nh}$$

Optimal bandwidth: Differentiate w.r.t. h and set to zero:

$$h^* = \left(\frac{\int K^2(u) du}{\sigma_K^4 \int (f''(x))^2 dx} \right)^{1/5} n^{-1/5}$$

- $h^* \propto n^{-1/5}$: bandwidth shrinks as n grows.
- At $h = h^*$: $\text{MISE}(\hat{f}_n, f) = O(n^{-4/5})$. (optimal rate!)
- Comparing with histogram: $n^{-4/5} \gg n^{-2/3}$ for large n .

h^* depends on $\int (f'')^2$, which is **unknown** — motivates cross-validation or normal reference rule.

KDE: Bandwidth Selection in Practice

Two practical methods:

1. Normal Reference Rule (Silverman's Rule of Thumb)

Assume f is approximately Gaussian. For a Gaussian kernel:

$$\hat{h} = 1.06 \hat{\sigma} n^{-1/5}, \quad \hat{\sigma} = \min \left\{ s, \frac{Q}{1.34} \right\}$$

where s = sample std dev, Q = interquartile range.

Simple but works well only for unimodal, approximately normal densities.

2. Leave-one-out Cross-Validation

$$\hat{J}(h) = \frac{1}{h^2 n^2} \sum_i \sum_j K^{(2)} \left(\frac{X_i - X_j}{h} \right) + \frac{2}{nh} K(0)$$

where $K^{(2)}(z) = \int K(z - y)K(y) dy$. **Choose** $\hat{h} = \operatorname{argmin}_h \hat{J}(h)$.

Asymptotically optimal; computationally heavier.

KDE: Consistency and Asymptotic Normality

Theorem 4 (Consistency (Wasserman 2006))

If f is continuous at x , $h_n \rightarrow 0$, and $nh_n \rightarrow \infty$, then $\hat{f}_n(x) \xrightarrow{P} f(x)$.

Theorem 5 (Asymptotic Normality)

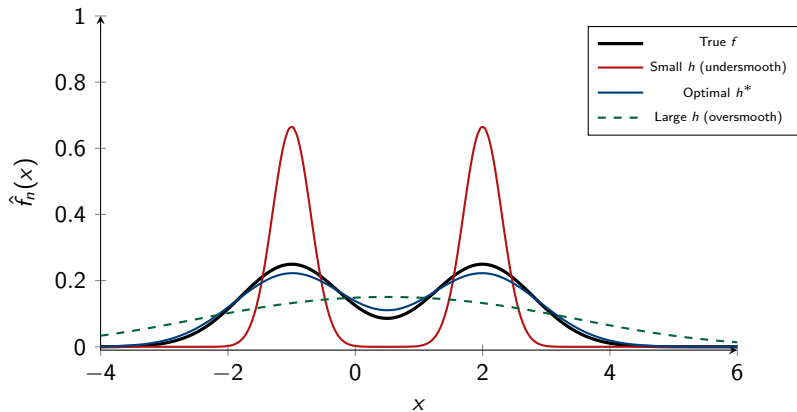
Under the conditions of Theorem 6.28 with $nh_n^5 \rightarrow \lambda$:

$$\sqrt{nh_n} \left[\hat{f}_n(x) - f(x) \right] \xrightarrow{d} \mathcal{N} \left(\frac{1}{2} \lambda^{1/2} \sigma_K^2 f''(x), f(x) \int K^2 \right).$$

Minimax lower bound (Tsybakov 2009, Thm. 2.1):

For $f \in \mathcal{H}(2, L)$, no estimator can converge faster than $n^{-4/5}$. KDE achieves this rate with optimal h^* : it is **rate-optimal**.

KDE: Effect of Bandwidth — Illustration



Small h : High variance, captures noise.

Optimal h^* : Best bias–variance balance.

Large h : High bias, oversmoothed (modes merged).

Projection Estimator: Motivation and Setup

Idea: Represent f as a series in an orthonormal basis, then *estimate the coefficients* from data.

Setup: Let $\{\varphi_j\}_{j \geq 1}$ be an orthonormal basis of $L^2([0, 1])$:

$$\int_0^1 \varphi_j(x) \varphi_k(x) dx = \delta_{jk}.$$

The true density has expansion:

$$f = \sum_{j=1}^{\infty} a_j \varphi_j, \quad a_j = \int_0^1 f(x) \varphi_j(x) dx = \mathbb{E}[\varphi_j(X)].$$

Plug-in estimator of Fourier coefficients:

$$\hat{a}_j = \frac{1}{n} \sum_{i=1}^n \varphi_j(X_i) \quad \Rightarrow \quad \mathbb{E}[\hat{a}_j] = a_j \quad (\text{unbiased!})$$

Projection Estimator: Definition and Bases

Definition 6 (Projection Estimator (Tsybakov 2009))

$$\hat{f}_D(x) = \sum_{j=1}^D \hat{a}_j \varphi_j(x) = \frac{1}{n} \sum_{i=1}^n \sum_{j=1}^D \varphi_j(X_i) \varphi_j(x)$$

where D is the **projection dimension** (smoothing parameter).

Key bases:

Trigonometric Basis

$\varphi_1 \equiv 1$,
 $\varphi_{2k}(x) = \sqrt{2} \cos(2\pi kx)$,
 $\varphi_{2k+1}(x) = \sqrt{2} \sin(2\pi kx)$.
 Best for smooth periodic f .

Histogram/Step Basis

$\varphi_j(x) = \sqrt{D} \mathbf{1}(x \in B_j)$,
 D equal bins on $[0, 1]$.
 Connects to histogram estimator.

Remark 2

The histogram estimator is a projection estimator onto the step-function basis.

Projection Estimator: Bias–Variance Analysis

Bias–variance decomposition of MISE:

$$\mathbb{E} \left[\|\hat{f}_D - f\|_2^2 \right] = \underbrace{\|f - f_D\|_2^2}_{\text{Bias}^2} + \underbrace{\mathbb{E} \left[\|\hat{f}_D - f_D\|_2^2 \right]}_{\text{Variance}}$$

where $f_D = \sum_{j=1}^D a_j \varphi_j$ is the best D -dimensional approximation.

Proposition 5 (Bias–Variance Bound)

$$\mathbb{E} \left[\|\hat{f}_D - f\|_2^2 \right] \leq \underbrace{\|f - f_D\|_2^2}_{\searrow \text{ as } D \uparrow} + \underbrace{\frac{\phi_0 D}{n}}_{\nearrow \text{ as } D \uparrow}$$

where $\phi_0 = \sup_{x,j} \varphi_j(x)^2$ (basis boundedness constant).

Bias $\propto D^{-2\beta}$ (for $f \in \mathcal{S}(\beta, L)$); Variance $\propto D/n$.

Optimal: $D^* = Cn^{1/(2\beta+1)} \Rightarrow \text{MISE} \propto n^{-2\beta/(2\beta+1)}$.

Projection Estimator: Model Selection (Oracle)

Goal: Choose D automatically from data (since β is unknown).

Oracle dimension:

$$D_{\text{oracle}} = \operatorname{argmin}_D \left\{ \|f - f_D\|_2^2 + \frac{\phi_0 D}{n} \right\}$$

– not computable (depends on unknown f). **Key trick:** Pythagoras gives $\|f - f_D\|_2^2 = \|f\|_2^2 - \|f_D\|_2^2$, so

$$D_{\text{oracle}} = \operatorname{argmin}_D \left\{ -\|f_D\|_2^2 + \frac{\phi_0 D}{n} \right\}.$$

Data-driven selection: Replace $\|f_D\|_2^2$ by $\|\hat{f}_D\|_2^2$:

$$\hat{D} = \operatorname{argmin}_{D \in \{1, \dots, n\}} \left\{ -\|\hat{f}_D\|_2^2 + \kappa \phi_0 \frac{D}{n} \right\}$$

where $\kappa > 0$ is a tuning constant.

Projection Estimator: Oracle Inequality

Theorem 6 (Oracle Inequality)

Consider nested models $S_1 \subset S_2 \subset \dots$ and let $\hat{f}_{\hat{D}}$ be the projection estimator with $\hat{D}$ selected by the data-driven criterion. If f is bounded, then there exist constants $C, C' > 0$ such that:

$$\mathbb{E} \left[\|\hat{f}_{\hat{D}} - f\|_2^2 \right] \leq C \inf_{D \in \{1, \dots, n\}} \left\{ \|f - f_D\|_2^2 + \frac{\phi_0 D}{n} \right\} + \frac{C'}{n}.$$

Interpretation:

- The data-driven estimator performs *as well as the oracle*, up to a constant C .
- The remainder $C'/n = o(n^{-2\beta/(2\beta+1)})$ is negligible.
- $\hat{f}_{\hat{D}}$ is an **adaptive estimator**: it automatically adapts to the unknown smoothness β .

Projection vs. Kernel: Summary Comparison

Property	Kernel (KDE)	Projection
Smoothing param.	Bandwidth h	Dimension D
Optimal rate	$n^{-4/5}$	$n^{-2\beta/(2\beta+1)}$
Adaptive selection	Cross-validation	Model selection
Output	Smooth function	Series expansion
Basis	Translation-invariant	Orthonormal basis
Computation	$O(n^2)$ (naive)	$O(nD)$
Boundary effects	Yes (need correction)	Depends on basis

Both methods achieve **minimax optimal rates**. The key difference is in their smoothing mechanism and adaptivity properties.

Chapter 2: Summary

- **Density estimation** reconstructs f from i.i.d. data $X_1, \dots, X_n$ with minimal assumptions.
- **Risk** (MISE/MSE) decomposes as $Bias^2 + Variance$; these trade off against the smoothing parameter.
- **Smoothness classes** ($\mathcal{H}(\beta, L)$, $\mathcal{S}(\beta, L)$) control bias: larger $\beta \Rightarrow$ faster rates.
- **Histogram**: $MISE \propto n^{-2/3}$; optimal $h^* \propto n^{-1/3}$. Simple but discontinuous and suboptimal.
- **KDE**: $MISE \propto n^{-4/5}$; optimal $h^* \propto n^{-1/5}$. Smooth, flexible, minimax optimal.
- **Projection**: Expand f in orthonormal basis; choose D via oracle inequality / model selection. Adaptive to unknown β .
- **Smoothing parameter selection**:
 - ▶ Cross-validation: asymptotically optimal, data-driven.
 - ▶ Normal reference rule: fast but assumes smooth unimodal f .
 - ▶ Oracle/adaptive selection: strong theoretical guarantees.

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